

LLM-note

Lecture notes on the theory of neural networks

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Contents

1	Three Questions in Deep-Learning Theory	2
1.1	Introduction	2
1.1.1	Definition of Neural Networks	3
1.1.2	How Neural Networks Are Used	4
1.1.3	Three Main Topics	5
1.2	Topic I: Scaling Laws and Large-Network Limits	5
1.2.1	Basic Questions	5
1.2.2	Empirical Scaling Laws	6
1.2.3	Optimization, Approximation, and Data Errors	6
1.2.4	Theoretical Approaches	7
1.2.5	Important Distinction: Scaling Is Not Just “More Parameters”	8
1.3	Topic II: Feature Learning and Representation Learning	9
1.3.1	Linear Models	9
1.3.2	Neural Networks Learn the Features	9
1.3.3	The Central Feature-Learning Question	10
1.3.4	A Minimal Two-Layer Model	10
1.3.5	What Is a Semantic Feature?	11
1.3.6	Distributed Representations and Superposition	11
1.3.7	Feature Geometry	12
1.3.8	Transformers and In-Context Learning	12
1.4	Topic III: Diffusion Models	13
1.4.1	Generative Modeling	13
1.4.2	Forward Diffusion Process	13
1.4.3	The Score Function	14
1.4.4	Why Is the Score Enough?	14
1.4.5	Learning the Score	14
1.4.6	Equivalent Noise-Prediction Formulation	15
1.4.7	Probability-Flow ODE	15
1.4.8	Geometrical Picture	16
1.4.9	Main Questions for Diffusion Models	16
1.5	Relations Between the Three Topics	16
1.5.1	Scaling and Feature Learning	16
1.5.2	Feature Learning and Diffusion	17
1.5.3	Scaling and Generative Modeling	17
1.6	Big Picture	17

2	Scaling Limits of Neural Networks	18
2.1	Basic setup	18
2.2	The scaling question	19
2.3	Depth versus width: deep linear networks	20
2.3.1	Non-commuting limits	21
2.4	Large-depth limits of ResNets	22
2.4.1	The $1/\sqrt{L}$ limit: Neural SDE	23
2.4.2	The $1/L$ limit: Neural ODE	23
2.4.3	What lies between $1/\sqrt{L}$ and $1/L$?	24
2.5	Number of parameters versus number of data points	24
2.5.1	High-dimensional limit and the Marchenko–Pastur law	25
2.5.2	Prediction risk and the optimal ridge parameter	26
2.6	Summary: the recurring idea	27
3	Infinite-Width Training Dynamics	29
3.1	Setup	29
3.2	Main Question	30
3.3	NTK Regime	30
3.4	Derivation of NTK Dynamics	31
3.5	Why Does the NTK Freeze?	32
3.6	Mean-Field Regime	32
3.7	Dynamics of an Individual Neuron	33
3.8	Evolution of the Neuron Distribution	34
3.9	NTK vs. Mean Field	35

Chapter 1

Three Questions in Deep-Learning Theory

Key ideas

- A neural network is a parametrized family of functions; what it learns is fixed jointly by the architecture, the data, and the optimization algorithm.
- **Scaling:** which simple effective theory emerges as $N, n, L, W, C \rightarrow \infty$ (NTK, mean field, DMFT, Gaussian processes), and why these limits need not commute.
- **Feature learning:** a network learns the feature map $\Phi_\theta(x)$ itself, not only its coefficients; features are directions in activation space and can be stored in superposition.
- **Diffusion:** learning the score $\nabla_x \log \rho_t$ by denoising is enough to reverse a noising process and sample from ρ_{data} .

1.1 Introduction

A useful viewpoint is that a neural network is a *parametrized family of functions*. We choose an architecture, which determines a family

$$\{f_\theta : \mathcal{X} \rightarrow \mathcal{Y}\}_{\theta \in \Theta}, \quad (1.1)$$

and learning consists of selecting the parameter θ using data.

Thus there are three conceptually distinct ingredients:

1. the **architecture**, which specifies the function class;
2. the **data**, which provide information about the unknown target;
3. the **optimization algorithm**, which determines which member of the function class is actually learned.

One of the central questions in the theory of neural networks is therefore

$$\boxed{\text{Architecture} + \text{Data} + \text{Optimization} \implies \text{Learned function / representation.}} \quad (1.2)$$

1.1.1 Definition of Neural Networks

1.1.1.1 Fully connected neural networks

Consider an input

$$x \in \mathbb{R}^{N_0}. \quad (1.3)$$

Define

$$z^{(0)}(x) = x. \quad (1.4)$$

A fully connected neural network with $L + 1$ layers is recursively defined by

$$z^{(\ell+1)}(x) = W^{(\ell+1)}\phi\left(z^{(\ell)}(x)\right) + b^{(\ell+1)}, \quad \ell = 0, \dots, L, \quad (1.5)$$

where

$$W^{(\ell+1)} \in \mathbb{R}^{N_{\ell+1} \times N_\ell}, \quad b^{(\ell+1)} \in \mathbb{R}^{N_{\ell+1}}, \quad (1.6)$$

and ϕ is a nonlinear activation function applied componentwise.¹

Typical choices include

$$\text{ReLU}(x) = \max(0, x), \quad (1.7)$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad (1.8)$$

$$\text{GELU}(x) = x\Phi(x), \quad (1.9)$$

where Φ is the standard Gaussian cumulative distribution function.

The output of the network is

$$f_\theta(x) \equiv z^{(L+1)}(x), \quad (1.10)$$

where the parameter set is

$$\theta = \left\{ W^{(1)}, \dots, W^{(L+1)}, b^{(1)}, \dots, b^{(L+1)} \right\}. \quad (1.11)$$

The nonlinear activation is essential. Without ϕ , a composition of linear layers would still be only a linear transformation:

$$W^{(L)}W^{(L-1)} \dots W^{(1)}x = W_{\text{eff}}x. \quad (1.12)$$

1.1.1.2 Residual neural networks

Instead of replacing the current representation at every layer, a residual network learns a correction:

$$z^{(\ell+1)} = z^{(\ell)} + F_{\theta_\ell}\left(z^{(\ell)}\right). \quad (1.13)$$

A typical residual block is

$$F_{\theta_\ell}(z) = W_2^{(\ell)}\phi\left(W_1^{(\ell)}z + b_1^{(\ell)}\right) + b_2^{(\ell)}. \quad (1.14)$$

Therefore,

$$z^{(\ell+1)} = z^{(\ell)} + W_2^{(\ell)}\phi\left(W_1^{(\ell)}z^{(\ell)} + b_1^{(\ell)}\right) + b_2^{(\ell)}. \quad (1.15)$$

¹With this convention ϕ also acts on the raw input $z^{(0)} = x$. Lecture 2 uses the equally common convention $z^{(1)} = W^{(1)}x + b^{(1)}$; nothing below depends on the choice.

The important physical picture is that each layer makes only an incremental update,

$$\Delta z^{(\ell)} = F_{\theta_\ell}(z^{(\ell)}). \quad (1.16)$$

In a formal continuum-depth limit, a residual network resembles a dynamical system,

$$\frac{dz(t)}{dt} = F_{\theta(t)}(z(t)), \quad (1.17)$$

which motivates the connection between residual networks and differential equations.

1.1.2 How Neural Networks Are Used

Suppose there is an unknown target function

$$f : \mathcal{X} \rightarrow \mathcal{Y}. \quad (1.18)$$

We observe a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad y_i = f(x_i), \quad (1.19)$$

where n denotes the number of training samples.

The standard supervised-learning procedure is:

1. **Data:** observe

$$\{(x_i, f(x_i))\}_{i=1}^n. \quad (1.20)$$

The goal is to infer the unknown function f from finitely many observations.

2. **Initialization:** choose an initial parameter

$$\theta_0 \sim P_{\text{init}}. \quad (1.21)$$

3. **Optimization:** minimize a loss function. For regression, a common choice is

$$\mathcal{L}(\theta) = \frac{1}{2n} \sum_{i=1}^n \|f(x_i) - f_\theta(x_i)\|^2. \quad (1.22)$$

Gradient descent gives

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t), \quad (1.23)$$

where η is the learning rate.

In practice one usually uses stochastic gradient descent,

$$\theta_{t+1} = \theta_t - \eta \widehat{\nabla_\theta \mathcal{L}}, \quad (1.24)$$

where the gradient is estimated from a minibatch.

4. **Generalization / test:** for a new point x_{new} that was not in the training set, ask whether

$$f(x_{\text{new}}) \approx f_{\theta_{\text{learned}}}(x_{\text{new}}). \quad (1.25)$$

A neural network is therefore not useful merely because it can fit the training data. The central problem is whether it can *generalize*.

1.1.3 Three Main Topics

We focus on three broad theoretical questions:

1. **Scaling laws and large-network limits:** What happens as the model size, dataset size, and computational budget become very large?
2. **Feature learning and representation learning:** What internal representations does the neural network learn, and why do these representations help generalization?
3. **Generative models, especially diffusion models:** How can a neural network learn an entire high-dimensional probability distribution and generate new samples from it?

1.2 Topic I: Scaling Laws and Large-Network Limits

1.2.1 Basic Questions

Modern neural networks involve several potentially large parameters:

$$N = \text{number of model parameters}, \quad (1.26)$$

$$n = \text{number of training samples}, \quad (1.27)$$

$$L = \text{depth}, \quad (1.28)$$

$$W = \text{width}, \quad (1.29)$$

$$C = \text{computational budget}. \quad (1.30)$$

The theoretical problem resembles statistical mechanics: instead of studying one finite neural network in detail, we would like to identify simple laws that emerge in appropriate large-scale limits.

Some central questions are:

1. What happens when

$$N, n, L, W, C \rightarrow \infty? \quad (1.31)$$

These limits need not commute. For example,

$$\lim_{n \rightarrow \infty} \lim_{W \rightarrow \infty} \quad (1.32)$$

may describe a different regime from

$$\lim_{W \rightarrow \infty} \lim_{n \rightarrow \infty} . \quad (1.33)$$

Therefore the interesting object is often a *joint scaling limit*, analogous to a thermodynamic limit.

2. Does a complicated neural network reduce to a simpler effective theory in such a limit?
3. Can performance at very large scale be predicted from experiments at relatively small scale?

4. Which resource is the bottleneck:

$$\text{model size,} \quad \text{data,} \quad \text{optimization,} \quad \text{compute?} \quad (1.34)$$

5. When does increasing the model size actually improve performance, and when does performance saturate?

1.2.2 Empirical Scaling Laws

Empirically, many machine-learning systems approximately obey power-law scaling over a substantial range of scales.

For example, test loss may behave as

$$\boxed{\mathcal{L}(C) = \mathcal{L}_\infty + AC^{-\beta}}, \quad (1.35)$$

where

$$C = \text{compute}, \quad (1.36)$$

$\beta > 0$ is a scaling exponent, and $\mathcal{L}_\infty$ is an irreducible loss floor.

Similarly one may study

$$\mathcal{L}(N) \approx \mathcal{L}_\infty + A_N N^{-\alpha_N}, \quad (1.37)$$

$$\mathcal{L}(n) \approx \mathcal{L}_\infty + A_n n^{-\alpha_n}. \quad (1.38)$$

More generally,

$$\mathcal{L} = \mathcal{L}(N, n, C, \dots). \quad (1.39)$$

The interesting theoretical questions are not only whether these power laws exist, but also:

1. why power laws emerge;
2. whether the exponents are universal;
3. how the exponents depend on the data distribution and architecture;
4. how to optimally allocate a fixed computational budget between increasing N , increasing n , and training for longer.

1.2.3 Optimization, Approximation, and Data Errors

It is useful to conceptually decompose the total error into several pieces:

$$\mathcal{E}_{\text{total}} \sim \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{estimation}} + \mathcal{E}_{\text{optimization}}. \quad (1.40)$$

Here:

- $\mathcal{E}_{\text{approx}}$ measures whether the architecture is expressive enough to represent the desired function;
- $\mathcal{E}_{\text{estimation}}$ arises because the training dataset is finite;
- $\mathcal{E}_{\text{optimization}}$ measures whether the optimizer succeeds in finding a good parameter configuration.

Scaling the number of parameters only directly attacks the first of these. Therefore increasing model size alone need not always improve performance.

1.2.4 Theoretical Approaches

1.2.4.1 Neural Tangent Kernel

Consider the network output

$$f_\theta(x). \quad (1.41)$$

Linearize it around the initialization θ_0 :

$$f_\theta(x) \approx f_{\theta_0}(x) + \nabla_\theta f_{\theta_0}(x)^T (\theta - \theta_0). \quad (1.42)$$

Define the Neural Tangent Kernel

$$K_{\text{NTK}}(x, x') = \nabla_\theta f_{\theta_0}(x)^T \nabla_\theta f_{\theta_0}(x'). \quad (1.43)$$

In an appropriate infinite-width limit, the NTK becomes approximately deterministic and changes very little during training.

The nonlinear neural-network training problem is then reduced to an effective kernel regression problem.

The physical picture is

$$\boxed{\text{very wide network} \Rightarrow \text{approximately fixed features} + \text{linearized dynamics.}} \quad (1.44)$$

This regime is often called the *lazy-training regime*.

A major limitation is that realistic neural networks often exhibit substantial *feature learning*, whereas the simplest NTK limit freezes the features.

1.2.4.2 Mean-Field Theory

Another large-width limit treats individual neurons as interacting particles.

Consider a two-layer network

$$f_m(x) = \frac{1}{m} \sum_{j=1}^m a_j \sigma(w_j^T x). \quad (1.45)$$

Instead of following every neuron separately, define an empirical distribution of neuron parameters,

$$\rho_m(a, w) = \frac{1}{m} \sum_{j=1}^m \delta(a - a_j) \delta(w - w_j). \quad (1.46)$$

As

$$m \rightarrow \infty, \quad (1.47)$$

one can sometimes derive a deterministic evolution equation for

$$\rho(a, w, t). \quad (1.48)$$

Schematically,

$$\partial_t \rho = \mathcal{F}[\rho]. \quad (1.49)$$

This is analogous to the passage

$$\text{many particles} \rightarrow \text{continuum density} \quad (1.50)$$

in statistical mechanics.

Unlike the simplest NTK regime, the distribution of features w can evolve, so mean-field theory can describe genuine feature learning.

1.2.4.3 Dynamical Mean-Field Theory

For sufficiently complicated random neural networks, especially deep networks or recurrent systems, one can seek an effective stochastic description of a typical neuron or feature.

The philosophy is analogous to dynamical mean-field theory in many-body physics:

$$\boxed{\text{high-dimensional interacting system} \rightarrow \text{self-consistent effective single-site dynamics.}} \quad (1.51)$$

Instead of tracking every neuron explicitly, one derives self-consistency equations for quantities such as

$$C(t, t') = \langle z(t)z(t') \rangle, \quad (1.52)$$

$$R(t, t') = \frac{\delta \langle z(t) \rangle}{\delta h(t')}. \quad (1.53)$$

These quantities play roles analogous to correlation and response functions in statistical physics.

1.2.4.4 Bayesian Inference

A different viewpoint is to place a prior distribution on the network parameters,

$$P(\theta), \quad (1.54)$$

and use Bayes' rule:

$$P(\theta | \mathcal{D}) = \frac{P(\mathcal{D} | \theta)P(\theta)}{P(\mathcal{D})}. \quad (1.55)$$

Predictions are then obtained by averaging over the posterior:

$$P(y_* | x_*, \mathcal{D}) = \int d\theta P(y_* | x_*, \theta)P(\theta | \mathcal{D}). \quad (1.56)$$

For some infinitely wide neural networks, the prior over functions becomes a Gaussian process,

$$f(x) \sim \text{GP}(0, K(x, x')). \quad (1.57)$$

This provides another analytically tractable infinite-width limit.

1.2.5 Important Distinction: Scaling Is Not Just “More Parameters”

When discussing scaling, one should distinguish several directions:

$$\text{model scaling : } N \rightarrow \infty, \quad (1.58)$$

$$\text{data scaling : } n \rightarrow \infty, \quad (1.59)$$

$$\text{compute scaling : } C \rightarrow \infty, \quad (1.60)$$

$$\text{context / horizon scaling : } T \rightarrow \infty. \quad (1.61)$$

Good scaling often requires these quantities to grow together.

Thus,

$$\boxed{\text{Scaling up the model} \neq \text{scaling up the learning problem.}} \quad (1.62)$$

1.3 Topic II: Feature Learning and Representation Learning

1.3.1 Linear Models

Before studying neural networks, consider a linear model in a fixed feature space:

$$f_{\theta}(x) = \theta^T \Phi(x) = \sum_{j=1}^M \theta_j \varphi_j(x), \quad (1.63)$$

where

$$\Phi(x) = \begin{pmatrix} \varphi_1(x) \\ \vdots \\ \varphi_M(x) \end{pmatrix} \quad (1.64)$$

is a feature map.

The important point is that the features

$$\varphi_j(x) \quad (1.65)$$

are fixed. Learning only determines their coefficients θ_j .

Therefore a linear model solves

fixed representation + learned linear coefficients.

(1.66)

1.3.2 Neural Networks Learn the Features

Consider the final layer of a neural network:

$$f_i(x; \theta) = \sum_{j=1}^{N_L} W_{ij}^{(L+1)} \phi_j(x; \theta), \quad (1.67)$$

where

$$\phi_j(x; \theta) = \phi\left(z_j^{(L)}(x; \theta)\right). \quad (1.68)$$

From the perspective of the final layer, this looks exactly like a linear model:

$$f(x) = W^{(L+1)} \Phi_{\theta}(x). \quad (1.69)$$

The crucial difference is

$\Phi_{\theta}(x)$ itself depends on the learned parameters.

(1.70)

Thus neural-network training does two things simultaneously:

1. it learns how to combine features;
2. it learns what the features themselves should be.

This is the basic meaning of *representation learning*.

1.3.3 The Central Feature-Learning Question

The basic theoretical problem is:

$$\boxed{\text{Data} + \text{Architecture} + \text{Optimization} + \text{Initialization} \rightarrow \text{Learned features.}} \quad (1.71)$$

More explicitly, we would like to understand

$$\Phi_{\theta_t}(x) \quad (1.72)$$

as a function of training time t .

Questions include:

1. Which features emerge first during training?
2. Which properties of the data determine the learned representation?
3. Why are some features reused across many tasks?
4. When are representations sparse or distributed?
5. When does the network discover semantic concepts?
6. How do learned representations change with width and depth?

1.3.4 A Minimal Two-Layer Model

Consider

$$f(x; t) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j(t) \sigma(w_j(t)^T x). \quad (1.73)$$

There are two qualitatively different kinds of learning.

If only

$$a_j(t) \quad (1.74)$$

change while

$$w_j(t) \approx w_j(0), \quad (1.75)$$

then the network is approximately learning a linear combination of fixed features

$$\sigma(w_j(0)^T x). \quad (1.76)$$

By contrast, if

$$w_j(t) \quad (1.77)$$

move substantially, the features themselves change:

$$\sigma(w_j(0)^T x) \longrightarrow \sigma(w_j(t)^T x). \quad (1.78)$$

This is genuine feature learning.

This also makes the connection with Topic I clear:

$$\boxed{\text{NTK} / \text{lazy limit} \approx \text{weak feature learning,}} \quad (1.79)$$

whereas other scaling limits can preserve nontrivial feature dynamics.

Remark (parametrization). The prefactor $1/\sqrt{m}$ used here is the NTK scaling, while the mean-field model of Section 1.2.4.2 uses $1/m$. With $1/\sqrt{m}$ and an $O(1)$ learning rate, each w_j moves by only $O(1/\sqrt{m})$ during training, so the features freeze as $m \rightarrow \infty$. With $1/m$ and a learning rate scaled up by m , each w_j moves by $O(1)$. The choice of normalization is therefore what selects lazy versus feature-learning behavior; Lecture 3 derives both limits.

1.3.5 What Is a Semantic Feature?

Suppose a vision model is given an image of the Golden Gate Bridge.

At the raw-input level, the image consists only of pixel values:

$$x = (x_1, x_2, \dots, x_N). \quad (1.80)$$

A low-level feature might detect:

$$\text{edges, colors, textures.} \quad (1.81)$$

A higher-level representation may encode concepts such as

$$\text{bridge, suspension bridge, Golden Gate Bridge, San Francisco.} \quad (1.82)$$

These are called *semantic features* because they correspond to meaningful concepts rather than simple local properties of the input.

An important question is whether such concepts correspond to individual neurons or to directions in a high-dimensional representation space.

1.3.6 Distributed Representations and Superposition

A naive picture is

$$\text{one neuron} \leftrightarrow \text{one concept.} \quad (1.83)$$

However, modern neural networks often appear to use distributed representations.

A feature may correspond to a direction

$$v_\alpha \in \mathbb{R}^d \quad (1.84)$$

in activation space, rather than to a single neuron.

An activation vector may schematically be written as

$$h(x) \approx \sum_{\alpha} c_{\alpha}(x) v_{\alpha}. \quad (1.85)$$

If the number of useful features is larger than the activation dimension, the network may store multiple features in non-orthogonal directions:

$$v_{\alpha}^T v_{\beta} \neq 0. \quad (1.86)$$

This phenomenon is often called *superposition*.

The rough idea is

$$\boxed{\text{number of useful features} > \text{number of available neurons}} \quad (1.87)$$

so the network compresses many features into a smaller-dimensional space.

This can lead to *polysemantic* neurons, where one neuron participates in representing several apparently unrelated concepts.

1.3.7 Feature Geometry

Rather than analyzing individual neurons, it is often more useful to study the geometry of representation space.

Let

$$h_\theta(x) \in \mathbb{R}^d \quad (1.88)$$

denote the learned representation.

Then one may study distances

$$d_\theta(x, x') = \|h_\theta(x) - h_\theta(x')\|, \quad (1.89)$$

similarities

$$K_\theta(x, x') = h_\theta(x)^T h_\theta(x'), \quad (1.90)$$

or covariance matrices

$$C = \mathbb{E}_x [h_\theta(x) h_\theta(x)^T]. \quad (1.91)$$

The spectrum of C provides information about the effective dimensionality of the learned representation.

1.3.8 Transformers and In-Context Learning

Feature learning becomes particularly interesting in transformers.

A transformer maps a sequence

$$x_1, \dots, x_T \quad (1.92)$$

to internal representations

$$h_1^{(\ell)}, \dots, h_T^{(\ell)}. \quad (1.93)$$

Self-attention performs

$$q_i = W_Q h_i, \quad (1.94)$$

$$k_j = W_K h_j, \quad (1.95)$$

$$v_j = W_V h_j, \quad (1.96)$$

followed by

$$\text{Attention}(i) = \sum_j A_{ij} v_j, \quad (1.97)$$

with

$$A_{ij} = \frac{\exp(q_i^T k_j / \sqrt{d})}{\sum_{j'} \exp(q_i^T k_{j'} / \sqrt{d})}. \quad (1.98)$$

Unlike a fixed feature map, the representation of one token depends on the other tokens in the context.

This raises the question of *in-context learning*: how can the forward pass of a trained transformer implement something that resembles a learning algorithm?

Schematically,

$$\text{context examples} \rightarrow \text{internal representation} \rightarrow \text{new prediction}. \quad (1.99)$$

One viewpoint is that training does not merely teach the model a fixed input-output function; it can teach the model an *algorithm for extracting a task from context*.

1.4 Topic III: Diffusion Models

1.4.1 Generative Modeling

Suppose the data are drawn from an unknown high-dimensional distribution

$$x_0 \sim \rho_0(x). \quad (1.100)$$

For example, x_0 may be an image.

The goal of a generative model is not merely to predict a label. Instead, we want to generate a new sample

$$x_{\text{new}} \sim \rho_0. \quad (1.101)$$

The main difficulty is that $\rho_0(x)$ can be an extremely complicated distribution in a very high-dimensional space.

1.4.2 Forward Diffusion Process

Diffusion models introduce a continuous family of distributions

$$\rho_t, \quad t \in [0, T], \quad (1.102)$$

such that

$$\rho_{t=0} = \rho_{\text{data}}, \quad (1.103)$$

while

$$\rho_T \approx \mathcal{N}(0, I). \quad (1.104)$$

The forward process gradually adds Gaussian noise.

A common parametrization is

$$\boxed{x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).} \quad (1.105)$$

Initially,

$$\alpha_0 \approx 1, \quad \sigma_0 \approx 0, \quad (1.106)$$

so

$$x_t \approx x_0. \quad (1.107)$$

At sufficiently large t ,

$$\alpha_T \approx 0, \quad \sigma_T \approx 1, \quad (1.108)$$

so

$$x_T \approx \epsilon \sim \mathcal{N}(0, I). \quad (1.109)$$

Thus,

$$\boxed{\text{complicated data distribution} \rightarrow \text{simple Gaussian distribution.}} \quad (1.110)$$

1.4.3 The Score Function

Define the score of the intermediate distribution ρ_t by

$$\boxed{s_t(x) = \nabla_x \log \rho_t(x)}. \quad (1.111)$$

The score is a vector field in data space.

Intuitively, $s_t(x)$ points toward directions in which the probability density increases most rapidly.

For a one-dimensional Gaussian,

$$\rho(x) \propto e^{-x^2/(2\sigma^2)}, \quad (1.112)$$

we have

$$s(x) = \frac{d}{dx} \log \rho(x) = -\frac{x}{\sigma^2}. \quad (1.113)$$

Therefore the score points back toward the high-probability region near $x = 0$.

1.4.4 Why Is the Score Enough?

Suppose the forward diffusion is described by the stochastic differential equation

$$dx = f(x, t) dt + g(t) dW_t. \quad (1.114)$$

The evolution of the probability density is described by a Fokker–Planck equation.

Remarkably, the corresponding reverse-time stochastic dynamics can be written in terms of the score:

$$dx = \left[f(x, t) - g(t)^2 \nabla_x \log \rho_t(x) \right] dt + g(t) d\bar{W}_t, \quad (1.115)$$

where the equation is integrated from $t = T$ back toward $t = 0$.

Therefore, if we know

$$s_t(x) = \nabla_x \log \rho_t(x) \quad (1.116)$$

for every t , we can reverse the noising process.

The central idea is consequently

$$\boxed{\text{learn the score} \implies \text{reverse diffusion} \implies \text{generate data.}} \quad (1.117)$$

1.4.5 Learning the Score

Introduce a neural network

$$s_\theta(x, t) \quad (1.118)$$

that approximates

$$\nabla_x \log \rho_t(x). \quad (1.119)$$

For the Gaussian corruption

$$x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad (1.120)$$

the conditional distribution is

$$p(x_t \mid x_0) = \mathcal{N}(\alpha_t x_0, \sigma_t^2 I). \quad (1.121)$$

Its score is known exactly:

$$\nabla_{x_t} \log p(x_t | x_0) = -\frac{x_t - \alpha_t x_0}{\sigma_t^2}. \quad (1.122)$$

Since

$$x_t - \alpha_t x_0 = \sigma_t \epsilon, \quad (1.123)$$

this becomes

$$\nabla_{x_t} \log p(x_t | x_0) = -\frac{\epsilon}{\sigma_t}. \quad (1.124)$$

Therefore one can train the network using denoising score matching:

$$\mathcal{L}_{\text{score}}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left[\lambda(t) \left\| s_\theta(x_t, t) + \frac{\epsilon}{\sigma_t} \right\|^2 \right]. \quad (1.125)$$

Thus the training problem can be reduced to a supervised regression problem with artificially generated Gaussian noise.

1.4.6 Equivalent Noise-Prediction Formulation

Instead of directly predicting the score, many diffusion models predict the noise

$$\epsilon_\theta(x_t, t). \quad (1.126)$$

Since

$$s_t(x_t) \sim -\frac{\epsilon}{\sigma_t}, \quad (1.127)$$

the two representations are directly related:

$$s_\theta(x_t, t) = -\frac{\epsilon_\theta(x_t, t)}{\sigma_t}. \quad (1.128)$$

The corresponding loss is approximately

$$\mathcal{L}_\epsilon = \mathbb{E} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]. \quad (1.129)$$

1.4.7 Probability-Flow ODE

The stochastic reverse dynamics are not the only way to generate samples.

The same family of marginal distributions ρ_t can also be reproduced by a deterministic probability-flow ODE,

$$\frac{dx}{dt} = f(x, t) - \frac{1}{2}g(t)^2 s_t(x). \quad (1.130)$$

Thus diffusion models can be understood either as

$$\text{stochastic reverse dynamics} \quad (1.131)$$

or as

$$\text{deterministic transport through probability space.} \quad (1.132)$$

This also provides a connection to flow-based generative models and flow matching.

1.4.8 Geometrical Picture

At large noise,

$$\rho_T(x) \approx \mathcal{N}(0, I), \quad (1.133)$$

which is simple.

As t decreases, the score field gradually guides samples toward the high-probability regions of the data distribution.

Schematically,

$$\text{Gaussian noise} \rightarrow \text{coarse structure} \rightarrow \text{objects} \rightarrow \text{fine details}. \quad (1.134)$$

In this sense, the neural network learns a time-dependent vector field

$$s_\theta(x, t) \quad (1.135)$$

that describes the geometry of the probability distribution at different noise scales.

1.4.9 Main Questions for Diffusion Models

Important theoretical questions include:

1. Why is the score easier to learn than the probability density itself?
2. Why does adding noise make the learning problem easier?
3. How accurately must the score be learned in order to generate accurate samples?
4. How does error accumulate during the reverse process?
5. Why do diffusion models scale so effectively with model size and data?
6. What representations are learned by the denoising network?
7. What is the relation between diffusion models, optimal transport, normalizing flows, and flow matching?
8. How many reverse-time steps are fundamentally necessary?
9. Which aspects of the data distribution are learned at large noise and which are learned only at small noise?

1.5 Relations Between the Three Topics

Although scaling laws, feature learning, and diffusion models initially look like separate topics, they are closely connected.

1.5.1 Scaling and Feature Learning

The infinite-width limit used in NTK theory can suppress feature learning:

$$\text{width} \rightarrow \infty \quad \Rightarrow \quad \text{features approximately frozen}. \quad (1.136)$$

Therefore an important question is which scaling limit retains the rich feature learning observed in practical neural networks.

This illustrates why different limits need not commute.

1.5.2 Feature Learning and Diffusion

A diffusion model is trained to estimate

$$s_t(x) = \nabla_x \log \rho_t(x) \quad (1.137)$$

at many noise levels.

To do this, the network must learn useful internal representations of the data.

At large t , these representations may encode coarse global structure, whereas at small t they must encode increasingly fine details.

Thus diffusion models provide a natural laboratory for studying how representations depend on scale.

1.5.3 Scaling and Generative Modeling

A generative model must represent an entire probability distribution rather than merely a decision boundary.

This raises particularly sharp scaling questions:

$$\text{How much model capacity and data are required to represent } \rho_{\text{data}}? \quad (1.138)$$

One may therefore study scaling laws such as

$$\mathcal{L}_{\text{gen}} = \mathcal{L}(N, n, C), \quad (1.139)$$

and ask whether the same scaling ideas observed in supervised learning also govern modern generative models.

1.6 Big Picture

The three topics can be summarized as follows.

Topic	Central Question
Scaling	What simple theory emerges when neural networks become very large?
Feature learning	What internal representation does training discover?
Diffusion	How can a neural network learn and sample from an entire distribution?

A common theme is the search for an effective theory of learning:

$$\boxed{\text{microscopic parameter dynamics} \longrightarrow \text{macroscopic laws of learning.}} \quad (1.140)$$

From a statistical-physics perspective, neural networks are particularly interesting because they are large interacting dynamical systems whose effective degrees of freedom, representations, and scaling laws are themselves learned from data.

Chapter 2

Scaling Limits of Neural Networks

Key ideas

- Large networks are controlled by a few *dimensionless ratios*, not by the raw sizes. A scaling limit is specified by the path to infinity together with how the hyperparameters scale.
- **Deep linear network:** $\log(\|z^{(L)}\|/\|x\|) \rightarrow \mathcal{N}(-\xi/2, \xi/2)$ with effective depth $\xi = L/N$; the width-first and depth-first limits differ.
- **ResNet:** $s_L \sim L^{-1/2}$ is critical for iid layers and gives a Neural SDE; $s_L \sim L^{-1}$ requires weights that vary smoothly across depth and gives a Neural ODE; correlated weights with Hurst exponent H interpolate via $s_L \sim L^{-H}$.
- **Ridge regression:** at $\gamma = d/n$ the Marchenko–Pastur law gives the exact asymptotic risk, minimized at $\lambda_* = \gamma/\text{SNR}$.

2.1 Basic setup

Definition (MLP). For an input

$$x \in \mathbb{R}^{N_0},$$

a multilayer perceptron is defined recursively by

$$z^{(1)}(x) = W^{(1)}x + b^{(1)} \in \mathbb{R}^{N_1}, \quad (2.1)$$

$$z^{(\ell+1)}(x) = W^{(\ell+1)}\phi\left(z^{(\ell)}(x)\right) + b^{(\ell+1)}, \quad \ell = 1, \dots, L-1. \quad (2.2)$$

Here N_ℓ is the width of layer ℓ , while L is the depth of the network. In the simplest constant-width setting,

$$N_1 = \dots = N_L = N.$$

Definition (ResNet). A residual network takes the form

$$z^{(\ell+1)} = z^{(\ell)} + s_L F^{(\ell)}\left(z^{(\ell)}\right), \quad (2.3)$$

where, for example,

$$F^{(\ell)}(z) = W_2^{(\ell)} \phi(W_1^{(\ell)} z + b_1^{(\ell)}) + b_2^{(\ell)}. \quad (2.4)$$

The factor s_L controls how the residual branch must scale as the depth L increases.

2.2 The scaling question

A central question is:

What limiting dynamics arise when the model size, data size, and training time all become large?

There are many possible large parameters.

Model scales.

- depth L ,
- width N ,
- number of attention heads H ,
- embedding dimension, number of parameters, etc.

Data and optimization scales.

- number of data points n ,
- batch size B ,
- number of optimization steps T .

It is not enough to simply say that all of these quantities go to infinity. One must specify the *path* to infinity and how the hyperparameters scale:

$$\mathcal{H} = \{\eta, \sigma, B, \lambda, \dots\} = \mathcal{H}(L, N, n, T, \dots). \quad (2.5)$$

Motivations.

1. Identify a small number of macroscopic variables that describe learning, analogous to thermodynamic variables in statistical physics. Examples include activation norms, feature covariances, kernel spectra, and training loss.
2. Identify parameterizations for which scaling is predictable: when the model becomes larger, one would like quantities such as the optimal learning rate or initialization scale to remain controlled. This is the idea behind *hyperparameter transfer*.

We first study three simple examples:

1. depth L versus width N in a fully connected network;
2. the large-depth limit of a ResNet;
3. parameter dimension d versus number of data points n .

2.3 Depth versus width: deep linear networks

We first remove all nonlinear complications and take

$$\phi(z) = z, \quad b^{(\ell)} = 0.$$

Then

$$z^{(L)}(x) = W^{(L)}W^{(L-1)} \dots W^{(1)}x. \quad (2.6)$$

Assume all hidden layers have width N , and initialize

$$W_{ij}^{(\ell)} \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{1}{N}\right). \quad (2.7)$$

The $1/N$ variance scaling is the natural random-matrix scaling: for a fixed vector v ,

$$\mathbb{E} \left[\|W^{(\ell)}v\|_2^2 \mid v \right] = \|v\|_2^2. \quad (2.8)$$

More precisely, rotational invariance gives

$$\frac{\|W^{(\ell)}v\|_2^2}{\|v\|_2^2} \stackrel{d}{=} \frac{1}{N} \chi_N^2. \quad (2.9)$$

Since different layers are independent,

$$\frac{\|z^{(L)}(x)\|_2}{\|x\|_2} \stackrel{d}{=} \prod_{\ell=1}^L \sqrt{\frac{\chi_{N,\ell}^2}{N}}, \quad (2.10)$$

where the $\chi_{N,\ell}^2$'s are independent.

Taking the logarithm turns the product into a sum:

$$\log \frac{\|z^{(L)}(x)\|_2}{\|x\|_2} = \frac{1}{2} \sum_{\ell=1}^L \log \left(\frac{\chi_{N,\ell}^2}{N} \right). \quad (2.11)$$

For large N ,

$$\mathbb{E} \left[\log \left(\frac{\chi_N^2}{N} \right) \right] = -\frac{1}{N} + O(N^{-2}), \quad (2.12)$$

$$\text{Var} \left[\log \left(\frac{\chi_N^2}{N} \right) \right] = \frac{2}{N} + O(N^{-2}). \quad (2.13)$$

Therefore,

$$\log \frac{\|z^{(L)}(x)\|_2}{\|x\|_2} \approx \mathcal{N}\left(-\frac{L}{2N}, \frac{L}{2N}\right). \quad (2.14)$$

This immediately identifies the natural dimensionless parameter

$$\boxed{\xi = \frac{L}{N}} \quad (2.15)$$

which acts as an *effective depth*.

If

$$L, N \rightarrow \infty, \quad \frac{L}{N} \rightarrow \xi,$$

then

$$\log \frac{\|z^{(L)}(x)\|_2}{\|x\|_2} \Rightarrow \mathcal{N}\left(-\frac{\xi}{2}, \frac{\xi}{2}\right). \quad (2.16)$$

Thus the norm becomes approximately log-normal.

Important subtlety. Although

$$\mathbb{E}\|z^{(L)}(x)\|_2^2 = \|x\|_2^2 \quad (2.17)$$

for every L , the *typical* norm decreases:

$$\mathbb{E} \left[\log \frac{\|z^{(L)}(x)\|_2}{\|x\|_2} \right] < 0.$$

The conserved second moment is maintained by increasingly rare realizations with large norms. This is the distinction between an averaged quantity and a typical realization.

2.3.1 Non-commuting limits

The two limits $N \rightarrow \infty$ and $L \rightarrow \infty$ do not commute.

1. Width first: free-probability (FP) regime. Take

$$N \rightarrow \infty \quad \text{with fixed } L.$$

Then

$$\frac{\chi_N^2}{N} \rightarrow 1$$

and hence, for a fixed input,

$$\frac{\|z^{(L)}(x)\|}{\|x\|} \rightarrow 1. \quad (2.18)$$

The product matrix itself is *not* the identity; rather, its large- N singular-value spectrum is naturally described using free probability.

2. Depth first: multiplicative ergodic theorem (MET) regime. For fixed N and

$$L \rightarrow \infty,$$

the logarithm of the norm is a sum of independent random increments. Therefore

$$\frac{1}{L} \log \frac{\|z^{(L)}(x)\|}{\|x\|} \rightarrow \frac{1}{2} \mathbb{E} \log \left(\frac{\chi_N^2}{N} \right) < 0. \quad (2.19)$$

This is the multiplicative-ergodic/Lyapunov-exponent picture.

3. Double-scaling regime. The interpolation between these limits is

$$N, L \rightarrow \infty, \quad \frac{L}{N} \rightarrow \xi. \quad (2.20)$$

Hence L/N , rather than L alone, is the natural effective depth.

2.4 Large-depth limits of ResNets

Consider first the simple linear residual model

$$z^{(\ell+1)} = \left(I + s_L W^{(\ell)} \right) z^{(\ell)}, \quad (2.21)$$

where

$$W_{ij}^{(\ell)} \sim \mathcal{N}\left(0, \frac{1}{N}\right)$$

independently across layers.

Conditioned on $z^{(\ell)}$,

$$\mathbb{E} \left[\|z^{(\ell+1)}\|^2 \mid z^{(\ell)} \right] = \|z^{(\ell)}\|^2 + s_L^2 \mathbb{E} \left[\|W^{(\ell)} z^{(\ell)}\|^2 \right] \quad (2.22)$$

$$= (1 + s_L^2) \|z^{(\ell)}\|^2. \quad (2.23)$$

Therefore

$$\mathbb{E} \|z^{(L)}\|^2 = (1 + s_L^2)^L \|x\|^2. \quad (2.24)$$

Let

$$s_L = L^{-\beta}. \quad (2.25)$$

Then

$$(1 + s_L^2)^L \sim \exp \left(L s_L^2 \right) = \exp \left(L^{1-2\beta} \right). \quad (2.26)$$

This gives three regimes:

$\beta < \frac{1}{2} \quad :$ exploding regime, $\beta = \frac{1}{2} \quad :$ non-trivial diffusive regime, $\beta > \frac{1}{2} \quad :$ identity regime for iid centered layers.	(2.27)
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Thus, for standard iid initialization,

$s_L \sim \frac{1}{\sqrt{L}}$	(2.28)
-------------------------------	--------

is the critical scaling.

2.4.1 The $1/\sqrt{L}$ limit: Neural SDE

The residual increments have size

$$\Delta z_\ell \sim \frac{1}{\sqrt{L}},$$

while there are L independent increments. Hence their cumulative fluctuation is order one:

$$\sqrt{L} \times \frac{1}{\sqrt{L}} = O(1). \quad (2.29)$$

This is exactly the central-limit/Brownian scaling.

Schematically,

$$z^{(\ell+1)} = z^{(\ell)} + \frac{1}{\sqrt{L}} G(z^{(\ell)}) \xi_\ell \quad (2.30)$$

converges to a stochastic differential equation

$$\boxed{dz_t = G(z_t) dB_t.} \quad (2.31)$$

Thus the natural continuous-time limit of an iid deep ResNet is generically a *Neural SDE*, rather than a Neural ODE.

2.4.2 The $1/L$ limit: Neural ODE

Now suppose instead that the residual vector field changes smoothly with layer number:

$$F^{(\ell)}(z) \approx f\left(\frac{\ell}{L}, z\right). \quad (2.32)$$

Take

$$s_L = \frac{1}{L}. \quad (2.33)$$

Then

$$z^{(\ell+1)} - z^{(\ell)} = \frac{1}{L} f\left(\frac{\ell}{L}, z^{(\ell)}\right). \quad (2.34)$$

Writing

$$t = \frac{\ell}{L}, \quad \Delta t = \frac{1}{L},$$

this is simply the Euler discretization

$$\frac{z^{(\ell+1)} - z^{(\ell)}}{\Delta t} = f(t, z^{(\ell)}). \quad (2.35)$$

Hence

$$\boxed{\frac{dz_t}{dt} = f(t, z_t).} \quad (2.36)$$

The role of increasing L is now to make the time discretization finer.

Key point. There is no contradiction between the Neural SDE and Neural ODE pictures. They correspond to different assumptions about the regularity of the weights across depth:

$$\boxed{\begin{array}{ll} \text{iid / rough across layers} & \implies s_L \sim L^{-1/2} \quad (\text{SDE}), \\ \text{smooth / correlated across layers} & \implies s_L \sim L^{-1} \quad (\text{ODE}). \end{array}} \quad (2.37)$$

In particular, using $s_L = 1/L$ together with iid, zero-mean residual blocks does *not* produce a non-trivial ODE: the random residuals simply average away and the network approaches the identity map.

2.4.3 What lies between $1/\sqrt{L}$ and $1/L$?

Suppose

$$s_L = L^{-\beta}, \quad \frac{1}{2} < \beta < 1. \quad (2.38)$$

If the layers remain iid, then

$$Ls_L^2 = L^{1-2\beta} \rightarrow 0, \quad (2.39)$$

so the fluctuations disappear and the network approaches the identity. Thus there is no new non-trivial iid limit between the SDE and ODE scalings.

However, an intermediate regime becomes possible if the weights are correlated across depth. A useful model is fractional Brownian motion. Let B_t^H have Hurst exponent

$$H \in \left(\frac{1}{2}, 1\right).$$

Its fluctuations over L increments scale as

$$L^H,$$

rather than $\sqrt{L}$. Therefore the corresponding critical residual scale is

$$\boxed{s_L \sim L^{-H}}. \quad (2.40)$$

The two familiar limits are recovered as

$$H = \frac{1}{2} \implies \text{Brownian/SDE scaling}, \quad (2.41)$$

$$H = 1 \implies \text{smooth/ODE scaling}. \quad (2.42)$$

Thus one can think schematically of

$$\boxed{\text{roughness of weights across depth} \leftrightarrow \text{critical depth scaling}}. \quad (2.43)$$

2.5 Number of parameters versus number of data points

We now study the simplest supervised-learning model in which a similar dimensionless ratio appears.

Consider linear regression

$$f_\theta(x) = \theta^\top x, \quad \theta \in \mathbb{R}^d, \quad (2.44)$$

with data

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n. \quad (2.45)$$

Assume the isotropic Gaussian model

$$x_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d), \quad (2.46)$$

$$y_i = x_i^\top \theta_0 + \epsilon_i, \quad (2.47)$$

$$\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2), \quad (2.48)$$

and use an isotropic random-effects model for the true parameter,

$$\theta_0 \sim \mathcal{N}\left(0, \frac{r^2}{d} I_d\right), \quad (2.49)$$

so that typically

$$\|\theta_0\|^2 \approx r^2.$$

The ridge estimator is

$$\hat{\theta}_\lambda = \arg \min_{\theta} \left\{ \frac{1}{n} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 \right\}. \quad (2.50)$$

Its closed-form solution is

$$\boxed{\hat{\theta}_\lambda = \left(\frac{X^\top X}{n} + \lambda I \right)^{-1} \frac{X^\top y}{n}.} \quad (2.51)$$

Therefore the central random object is the sample covariance matrix

$$S = \frac{X^\top X}{n}. \quad (2.52)$$

2.5.1 High-dimensional limit and the Marchenko–Pastur law

Take

$$n, d \rightarrow \infty, \quad \boxed{\gamma = \frac{d}{n}} \text{ fixed.} \quad (2.53)$$

The empirical eigenvalue distribution of

$$S = \frac{X^\top X}{n}$$

converges to the Marchenko–Pastur distribution

$$d\mu_\gamma(t) = \rho_\gamma(t) dt + \left(1 - \frac{1}{\gamma}\right)_+ \delta_0, \quad (2.54)$$

where

$$\rho_\gamma(t) = \frac{\sqrt{(\lambda_+ - t)(t - \lambda_-)}}{2\pi\gamma t} \mathbf{1}_{[\lambda_-, \lambda_+]}(t), \quad (2.55)$$

with

$$\boxed{\lambda_\pm = (1 \pm \sqrt{\gamma})^2.} \quad (2.56)$$

Thus the spectrum depends on d and n primarily through the single aspect ratio

$$\boxed{\gamma = d/n}.$$

This is analogous to the effective depth L/N in the previous example.

Example: $d/n = 3$. For

$$\gamma = 3,$$

we have

$$\lambda_- = (1 - \sqrt{3})^2 \approx 0.536, \quad (2.57)$$

$$\lambda_+ = (1 + \sqrt{3})^2 \approx 7.464. \quad (2.58)$$

Moreover,

$$1 - \frac{1}{\gamma} = \frac{2}{3}. \quad (2.59)$$

Hence $2/3$ of the eigenvalues are exactly zero in the large-system limit. This simply reflects the fact that when $d > n$,

$$\text{rank}(X^\top X) \leq n,$$

so there are $d - n$ unconstrained parameter directions.

2.5.2 Prediction risk and the optimal ridge parameter

For a fresh input

$$x_{\text{test}} \sim \mathcal{N}(0, I_d),$$

define the excess prediction risk

$$R(\lambda) = \mathbb{E} \left[\left(x_{\text{test}}^\top (\hat{\theta}_\lambda - \theta_0) \right)^2 \right]. \quad (2.60)$$

In the high-dimensional isotropic limit,

$$R(\lambda) = r^2 \int \frac{\lambda^2}{(t + \lambda)^2} d\mu_\gamma(t) + \sigma_\epsilon^2 \gamma \int \frac{t}{(t + \lambda)^2} d\mu_\gamma(t). \quad (2.61)$$

The two terms have a simple interpretation:

$$\text{bias} = r^2 \int \frac{\lambda^2}{(t + \lambda)^2} d\mu_\gamma(t), \quad (2.62)$$

$$\text{variance} = \sigma_\epsilon^2 \gamma \int \frac{t}{(t + \lambda)^2} d\mu_\gamma(t). \quad (2.63)$$

Differentiating,

$$\frac{dR}{d\lambda} = 2 \left(r^2 \lambda - \sigma_\epsilon^2 \gamma \right) \int \frac{t}{(t + \lambda)^3} d\mu_\gamma(t). \quad (2.64)$$

Therefore

$$\lambda_* = \frac{\sigma_\epsilon^2 \gamma}{r^2}. \quad (2.65)$$

Equivalently, defining

$$\text{SNR} = \frac{r^2}{\sigma_\epsilon^2}, \quad (2.66)$$

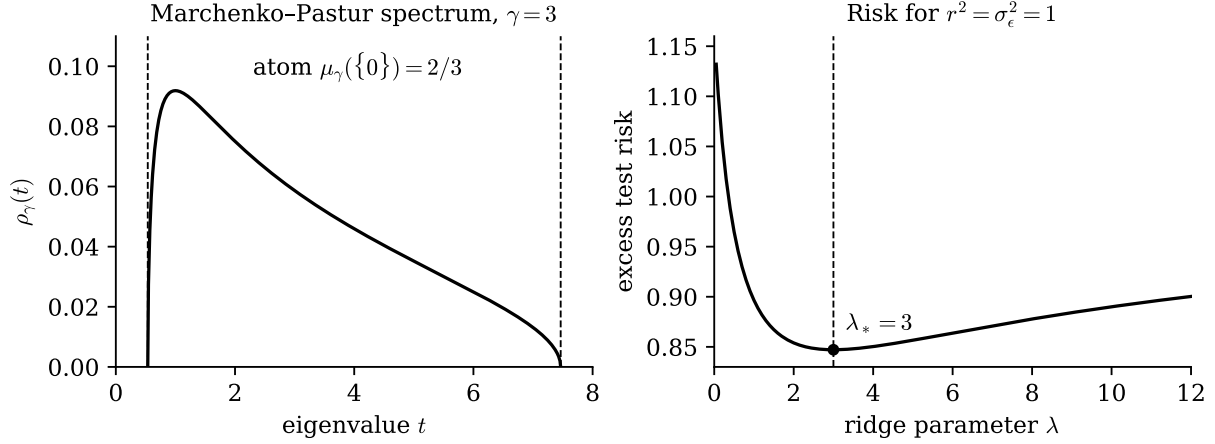


Figure 2.1: High-dimensional linear regression at aspect ratio $\gamma = d/n = 3$. Left: the continuous part of the Marchenko–Pastur spectrum; an additional atom of weight $2/3$ sits at zero. Right: the asymptotic excess prediction risk of ridge regression for $r^2 = \sigma_\epsilon^2 = 1$. The minimum occurs at $\lambda_* = \gamma\sigma_\epsilon^2/r^2 = 3$.

we obtain the simple scaling law

$$\boxed{\lambda_* = \frac{\gamma}{\text{SNR}}}. \quad (2.67)$$

For example, take

$$\gamma = 3, \quad r^2 = 1, \quad \sigma_\epsilon^2 = 1.$$

Then

$$\boxed{\lambda_* = 3}. \quad (2.68)$$

A numerical sweep of the Marchenko–Pastur prediction-risk formula gives the same minimum.

2.6 Summary: the recurring idea

These examples illustrate the same general principle: large systems are often controlled not by the raw sizes individually, but by a small number of *dimensionless ratios*.

deep linear network	: $\xi = L/N$,	
ResNet	: LS_L^2 (iid case),	
high-dimensional regression	: $\gamma = d/n$.	(2.69)

The goal of scaling theory is therefore to identify:

1. the correct macroscopic variables;
2. the dimensionless combinations that remain finite;
3. the corresponding limiting dynamics;
4. the scaling of hyperparameters required to obtain a non-trivial limit.

This is the basic philosophy behind studying neural networks through large- N , large- L , large-data, and large-training-time limits.

Chapter 3

Infinite-Width Training Dynamics

Key ideas

- The question is how the initialization and learning rate must scale with the width N for the limit $N \rightarrow \infty$ to exist and stay nontrivial. One family covers both answers: $f_\theta = \frac{1}{\gamma\sqrt{N}} \sum_i W_i^{(2)} \phi(W_i^{(1)} \cdot x / \sqrt{D})$ trained with learning rate $\eta\gamma^2$.
- **NTK** ($\gamma = 1$): each neuron moves only $O(N^{-1/2})$, the kernel K_θ freezes, and training is kernel regression, $\mathbf{r}(t) = e^{-\eta K_\infty t/n} \mathbf{r}(0)$.
- **Mean field** ($\gamma = \sqrt{N}$): each neuron moves $O(1)$ and feels the others only through their empirical distribution μ_t , which follows the Wasserstein gradient flow $\partial_t \mu_t = \eta \nabla \cdot (\mu_t \nabla \delta \mathcal{L} / \delta \mu_t)$.
- Lazy training versus feature learning is decided by the parametrization, not by the width alone.

3.1 Setup

Consider a two-layer neural network

$$f_\theta(x) = s_2 \sum_{i=1}^N W_i^{(2)} \phi(s_1 W_i^{(1)} \cdot x), \quad (3.1)$$

where $x \in \mathbb{R}^D$ is the input, N is the width, and

$$\theta_i = (W_i^{(1)}, W_i^{(2)})$$

denotes the parameters associated with neuron i .

The network can be viewed as a bipartite computational graph: the input is connected to N hidden neurons through $W_i^{(1)}$, and the hidden neurons are connected to the output through $W_i^{(2)}$.

We train the network by gradient descent,

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t), \quad (3.2)$$

with, for example, the squared loss

$$\mathcal{L}(\theta) = \frac{1}{2n} \sum_{\alpha=1}^n (f_{\theta}(x_{\alpha}) - y_{\alpha})^2. \quad (3.3)$$

Throughout this discussion, we first keep the number of data points n and the training time fixed, and study the limit

$$N \rightarrow \infty.$$

A more difficult problem is to study joint limits in which the number of data points or the training time also grows with N .

3.2 Main Question

The main question is:

How should the initialization and learning rate scale with the width N so that the limit $N \rightarrow \infty$ exists and remains nontrivial?

We take

$$W_{ij}^{(\ell)} \sim \mathcal{N}(0, 1), \quad (3.4)$$

and introduce the unified parameterization

$$f_{\theta}(x) = \frac{1}{\gamma\sqrt{N}} \sum_{i=1}^N W_i^{(2)} \phi\left(\frac{1}{\sqrt{D}} W_i^{(1)} \cdot x\right) \quad (3.5)$$

together with the learning-rate scaling

$$\theta_{t+1} = \theta_t - \eta\gamma^2 \nabla_{\theta} \mathcal{L}(\theta_t). \quad (3.6)$$

Two important regimes are

$$\gamma = 1 \quad \text{NTK / lazy-training regime,} \quad (3.7)$$

$$\gamma = \sqrt{N} \quad \text{mean-field regime.} \quad (3.8)$$

These two choices lead to qualitatively different infinite-width training dynamics.

3.3 NTK Regime

Take

$$\gamma = 1.$$

Then

$$f_{\theta}(x) = \frac{1}{\sqrt{N}} \sum_{i=1}^N W_i^{(2)} \phi\left(\frac{1}{\sqrt{D}} W_i^{(1)} \cdot x\right). \quad (3.9)$$

The important phenomenon is that, for large N , the parameters move only a small distance from their initialization. Therefore the network can be approximated by its first-order Taylor expansion around θ_0 :

$$\boxed{f_{\theta}^{\text{lin}}(x) = f_{\theta_0}(x) + \langle \theta - \theta_0, \nabla_{\theta} f_{\theta_0}(x) \rangle.} \quad (3.10)$$

Schematically, under suitable regularity assumptions and for finite training time,

$$\sup_{t \leq T} \|f_{\theta_t} - f_{\theta_t}^{\text{lin}}\| \longrightarrow 0, \quad (3.11)$$

$$\sup_{t \leq T} \|K_{\theta_t} - K_{\theta_0}\| \longrightarrow 0 \quad (3.12)$$

as $N \rightarrow \infty$.

Thus, in the NTK limit, training of the neural network becomes equivalent to training a linear model with the fixed feature map

$$x \mapsto \nabla_{\theta} f_{\theta_0}(x).$$

In this sense,

At infinite width, the nonlinear neural network has linearized training dynamics around its initialization.

3.4 Derivation of NTK Dynamics

It is convenient to consider continuous-time gradient flow,

$$\frac{d\theta_t}{dt} = -\eta \nabla_{\theta} \mathcal{L}(\theta_t). \quad (3.13)$$

Then

$$\frac{d}{dt} f_{\theta_t}(x) = \nabla_{\theta} f_{\theta_t}(x) \cdot \frac{d\theta_t}{dt} \quad (3.14)$$

$$= -\eta \nabla_{\theta} f_{\theta_t}(x) \cdot \nabla_{\theta} \mathcal{L}(\theta_t). \quad (3.15)$$

For the squared loss,

$$\nabla_{\theta} \mathcal{L} = \frac{1}{n} \sum_{\alpha=1}^n (f_{\theta}(x_{\alpha}) - y_{\alpha}) \nabla_{\theta} f_{\theta}(x_{\alpha}). \quad (3.16)$$

Therefore,

$$\boxed{\frac{d}{dt} f_{\theta_t}(x) = -\frac{\eta}{n} \sum_{\alpha=1}^n (f_{\theta_t}(x_{\alpha}) - y_{\alpha}) K_{\theta_t}(x, x_{\alpha}),} \quad (3.17)$$

where

$$\boxed{K_{\theta}(x, x') = \langle \nabla_{\theta} f_{\theta}(x), \nabla_{\theta} f_{\theta}(x') \rangle} \quad (3.18)$$

is the **Neural Tangent Kernel (NTK)**.

For the training data, define

$$f_{\alpha}(t) = f_{\theta_t}(x_{\alpha}), \quad r_{\alpha}(t) = f_{\alpha}(t) - y_{\alpha}.$$

Then

$$\dot{\mathbf{r}}(t) = -\frac{\eta}{n} K_{\theta_t} \mathbf{r}(t). \quad (3.19)$$

If the kernel becomes constant,

$$K_{\theta_t} \approx K_{\theta_0} \rightarrow K_{\infty},$$

then

$$\mathbf{r}(t) = \exp\left(-\frac{\eta}{n} K_{\infty} t\right) \mathbf{r}(0). \quad (3.20)$$

Thus the entire training dynamics is determined by a fixed kernel.

3.5 Why Does the NTK Freeze?

Because the network is a sum over neurons,

$$K_{\theta}(x, x') = \frac{1}{N} \sum_{i=1}^N K_{\theta_i}(x, x'), \quad (3.21)$$

so $K_{\theta} = O(1)$.

For a two-layer network, the Hessian with respect to neuron parameters has a block-diagonal structure,

$$\text{Hess}_{\theta} f = \text{diag} [\text{Hess}_{\theta_1} f, \dots, \text{Hess}_{\theta_N} f]. \quad (3.22)$$

In NTK parameterization, each neuron contributes only $O(N^{-1/2})$ to the network output. Correspondingly, each individual neuron moves only

$$\Delta\theta_i = O(N^{-1/2})$$

over an $O(1)$ training time.

Hence

$$\theta_i(t) \approx \theta_i(0),$$

and the features generated by each neuron barely change.

Therefore

$$\frac{d}{dt} K_{\theta_t} \longrightarrow 0 \quad \text{as } N \rightarrow \infty. \quad (3.23)$$

The nonlinear terms in the Taylor expansion vanish, leaving only the linearized dynamics.

$$\boxed{\text{NTK limit} \implies \text{kernel learning with nearly frozen features.}}$$

3.6 Mean-Field Regime

Now take

$$\gamma = \sqrt{N}.$$

Then

$$f_\theta(x) = \frac{1}{N} \sum_{i=1}^N W_i^{(2)} \phi\left(\frac{1}{\sqrt{D}} W_i^{(1)} \cdot x\right). \quad (3.24)$$

The learning-rate scaling becomes

$$\theta_{t+1} = \theta_t - \eta N \nabla_\theta \mathcal{L}(\theta_t). \quad (3.25)$$

Unlike the NTK regime, individual neurons now move by an $O(1)$ amount during training. Therefore the feature representation can evolve nontrivially.

Define

$$\vartheta_i = (W_i^{(1)}, W_i^{(2)}) \quad (3.26)$$

and the empirical distribution of neurons

$$\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{\vartheta_i(t)}. \quad (3.27)$$

Then the network output can be written as

$$f_{\mu_t^N}(x) = \int W^{(2)} \phi\left(\frac{1}{\sqrt{D}} W^{(1)} \cdot x\right) d\mu_t^N(W^{(1)}, W^{(2)}). \quad (3.28)$$

As

$$N \rightarrow \infty,$$

the discrete empirical measure μ_t^N approaches a continuous probability measure μ_t .

Thus instead of describing N individual neurons, we describe the evolution of the distribution of neurons.

3.7 Dynamics of an Individual Neuron

Define

$$\psi(\vartheta; x) = W^{(2)} \phi\left(\frac{1}{\sqrt{D}} W^{(1)} \cdot x\right). \quad (3.29)$$

Then

$$f_\mu(x) = \int \psi(\vartheta; x) d\mu(\vartheta).$$

The gradient of the network with respect to neuron i is

$$\nabla_{\vartheta_i} f_\theta(x_\alpha) = \frac{1}{N} \nabla_{\vartheta_i} \psi(\vartheta_i; x_\alpha). \quad (3.30)$$

Because the learning rate is scaled by N ,

$$\frac{d}{dt} \vartheta_i(t) = -\eta N \nabla_{\vartheta_i} \mathcal{L} \quad (3.31)$$

$$= -\frac{\eta}{n} \sum_{\alpha=1}^n (f_{\mu_t}(x_\alpha) - y_\alpha) \nabla_{\vartheta_i} \psi(\vartheta_i; x_\alpha). \quad (3.32)$$

Therefore,

$$\dot{\vartheta}_i = -\eta \mathcal{M}(\vartheta_i, \mu_t). \quad (3.33)$$

with

$$\mathcal{M}(\vartheta, \mu) = \frac{1}{n} \sum_{\alpha=1}^n (f_\mu(x_\alpha) - y_\alpha) \nabla_\vartheta \psi(\vartheta; x_\alpha), \quad (3.34)$$

which the next section identifies as $\nabla_\vartheta \delta \mathcal{L} / \delta \mu$.

The important point is that neuron i does not depend directly on all other neurons separately. It depends on them only through the empirical distribution

$$\mu_t.$$

This is a mean-field interaction.

3.8 Evolution of the Neuron Distribution

The loss can now be regarded as a functional of the measure,

$$\mathcal{L}[\mu] = \frac{1}{2n} \sum_{\alpha=1}^n (f_\mu(x_\alpha) - y_\alpha)^2. \quad (3.35)$$

Its functional derivative is

$$\frac{\delta \mathcal{L}}{\delta \mu}(\vartheta) = \frac{1}{n} \sum_{\alpha=1}^n (f_\mu(x_\alpha) - y_\alpha) \psi(\vartheta; x_\alpha). \quad (3.36)$$

Therefore the particle dynamics becomes

$$\dot{\vartheta} = -\eta \nabla_\vartheta \frac{\delta \mathcal{L}}{\delta \mu}(\vartheta). \quad (3.37)$$

If the particles follow a velocity field

$$v_t(\vartheta) = -\eta \nabla_\vartheta \frac{\delta \mathcal{L}}{\delta \mu_t}(\vartheta),$$

conservation of probability gives the continuity equation

$$\partial_t \mu_t + \nabla_\vartheta \cdot (\mu_t v_t) = 0. \quad (3.38)$$

Substituting v_t ,

$$\partial_t \mu_t = \eta \nabla_\vartheta \cdot \left(\mu_t \nabla_\vartheta \frac{\delta \mathcal{L}}{\delta \mu_t} \right). \quad (3.39)$$

This is a **Wasserstein gradient flow** of the loss functional $\mathcal{L}[\mu]$.

Indeed,

$$\frac{d}{dt} \mathcal{L}[\mu_t] = -\eta \int \left\| \nabla_\vartheta \frac{\delta \mathcal{L}}{\delta \mu_t} \right\|^2 d\mu_t(\vartheta) \leq 0. \quad (3.40)$$

Thus gradient descent on a finite collection of neurons becomes, at infinite width, gradient flow of a probability distribution in parameter space.

3.9 NTK vs. Mean Field

The two limits therefore have very different physical pictures:

$$\text{NTK:} \quad \theta_i(t) - \theta_i(0) \rightarrow 0, \quad (3.41)$$

$$K_{\theta_t} \approx K_{\theta_0}, \quad (3.42)$$

$$\text{features are approximately frozen;} \quad (3.43)$$

$$\text{Mean Field:} \quad \theta_i(t) - \theta_i(0) = O(1), \quad (3.44)$$

$$\mu_0 \rightarrow \mu_t, \quad (3.45)$$

$$\text{the neuron distribution and features evolve.} \quad (3.46)$$

Hence,

NTK limit	vs.	Mean-field limit
fixed kernel / lazy feature learning		distributional dynamics / feature learning.

The central conceptual change in the mean-field limit is

many interacting neurons	$\longrightarrow$	a probability measure evolving by optimal transport.
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